

PETROPAVL, Kazakhstan

WMO: 286760

Lat: 54.7923N

Lon: 69.1174E

Elev: 331

StdP: 14.52

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.1	-22.5	-33.4	0.8	-27.7	-28.0	1.2	-22.2	27.5	14.3	24.8	13.4	5.2	90	0.644

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.7	87.5	67.6	84.1	66.6	80.9	65.1	70.6	82.3	68.9	80.1	67.2	77.7	8.9	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.5	98.6	75.9	64.7	92.8	74.3	63.1	87.5	72.9	34.7	82.5	33.3	80.0	31.9	77.8	79.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.2	21.3	18.9	DB	-34.3	93.0	6.0	3.5	-38.6	95.6	-42.1	97.6	-45.5	99.6	-49.8	102.1
			WB	-34.2	73.8	5.9	2.4	-38.4	75.5	-41.9	76.9	-45.2	78.2	-49.4	79.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	36.4	0.9	4.6	19.6	40.1	55.2	64.5	66.7	63.3	52.4	39.4	20.3	7.4	(6)
(7)		DBStd	25.89	13.44	12.98	10.56	11.05	9.13	7.89	6.39	7.52	8.37	9.15	13.86	13.39	(7)
(8)		HDD50	6752	1523	1271	942	332	52	4	0	2	68	347	891	1320	(8)
(9)		HDD65	10754	1988	1691	1407	747	322	105	59	126	388	795	1341	1785	(9)
(10)		CDD50	1775	0	0	0	35	214	438	518	414	139	17	0	0	(10)
(11)		CDD65	301	0	0	0	0	19	89	111	73	9	0	0	0	(11)
(12)		CDH74	3034	0	0	0	12	319	933	986	670	113	1	0	0	(12)
(13)		CDH80	883	0	0	0	2	76	313	272	193	27	0	0	0	(13)
(14)	Wind	WSAvg	9.1	8.8	9.1	10.3	10.3	10.2	8.6	7.9	7.8	8.4	9.2	9.3	9.2	(14)
(15)	Precipitation	PrecAvg	14.5	0.8	0.6	0.6	0.7	1.2	1.7	2.4	2.0	1.1	1.2	1.1	0.8	(15)
(16)		PrecMax	23.4	1.7	1.9	1.6	1.8	3.0	4.4	8.2	4.8	3.0	3.8	2.7	2.2	(16)
(17)		PrecMin	9.7	0.0	0.0	0.0	0.0	0.0	0.3	0.4	0.1	0.2	0.1	0.0	0.0	(17)
(18)		PrecStd	3.1	0.4	0.4	0.4	0.5	0.7	0.9	1.5	1.0	0.7	0.8	0.6	0.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.7	34.1	45.5	74.8	87.0	91.3	91.3	89.4	83.8	68.9	46.6	34.1	(19)
(20)			MCWB	31.9	32.0	39.7	55.5	63.6	68.3	70.1	67.8	61.2	53.0	42.9	32.5	(20)
(21)		2%	DB	28.5	31.1	38.5	68.0	81.5	87.6	86.0	85.2	76.9	63.0	42.2	32.2	(21)
(22)			MCWB	27.5	29.7	35.1	52.3	60.2	66.7	68.9	67.6	58.9	50.8	39.7	30.8	(22)
(23)		5%	DB	25.3	27.8	36.1	62.5	77.4	83.6	82.6	81.3	71.1	57.6	39.2	29.6	(23)
(24)			MCWB	24.3	26.7	33.6	50.3	58.9	66.5	67.9	66.3	57.3	48.1	37.0	28.4	(24)
(25)		10%	DB	21.3	24.1	33.9	56.9	72.7	79.2	79.6	77.2	66.2	52.8	35.9	25.4	(25)
(26)			MCWB	20.5	22.8	32.0	47.7	56.5	64.6	66.6	64.3	55.1	46.1	34.0	24.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	31.9	32.4	39.8	57.5	66.1	71.9	73.0	72.0	64.0	55.4	43.2	33.1	(27)
(28)			MCDWB	32.6	33.9	44.5	70.2	81.4	85.4	84.1	84.5	77.0	65.2	45.5	33.6	(28)
(29)		2%	WB	27.5	29.7	35.6	53.7	62.9	69.7	70.9	69.6	60.7	52.1	40.0	30.9	(29)
(30)			MCDWB	28.2	30.8	37.7	65.2	77.6	82.6	81.9	80.7	72.6	60.4	41.8	32.0	(30)
(31)		5%	WB	24.3	26.7	33.9	50.9	60.3	67.5	69.3	67.6	58.3	48.9	37.1	28.2	(31)
(32)			MCDWB	25.3	27.7	35.8	61.3	74.0	79.9	79.8	78.0	68.9	56.1	39.0	29.6	(32)
(33)		10%	WB	20.4	22.8	32.0	47.9	57.9	65.5	67.7	65.6	56.0	46.3	34.2	24.2	(33)
(34)			MCDWB	21.3	24.0	33.8	56.4	70.2	76.8	77.6	75.2	64.7	51.9	35.6	25.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	14.1	16.2	16.1	17.4	21.5	20.2	18.7	19.4	18.5	14.5	11.7	13.5	(35)
(36)			MCDBR	14.3	14.5	13.6	25.9	28.6	25.9	23.7	25.4	26.4	23.8	11.9	12.4	(36)
(37)			MCWBR	14.1	13.8	11.1	13.5	13.4	11.6	10.6	11.4	13.0	14.0	10.0	12.0	(37)
(38)		5% WB	MCDBR	14.2	14.2	12.6	24.0	26.2	22.9	21.0	22.4	24.1	21.7	11.2	11.8	(38)
(39)			MCWBR	14.1	13.6	10.8	13.4	13.7	11.5	10.4	11.0	12.9	13.5	9.9	11.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.286	0.320	0.376	0.415	0.409	0.411	0.433	0.437	0.381	0.359	0.323	0.289	(40)	
(41)		taud	2.047	2.010	1.942	2.103	2.221	2.290	2.219	2.178	2.293	2.226	2.109	2.019	(41)	
(42)		Ebn at Noon	194	231	243	254	265	266	256	246	246	218	177	160	(42)	
(43)		Edh at Noon	22	33	44	44	41	39	41	41	32	27	21	18	(43)	
(44)	All-Sky Solar Radiation	RadAvg	316	657	1145	1458	1773	1905	1778	1441	997	511	250	200	(44)	
(45)		RadStd	23	32	93	151	138	145	131	129	78	79	34	28	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page